

Std - X

Sub - Science

CHAPTER 1

Chemical Reactions and Equations



Types of Changes

Physical Change

Temporary Change

No new substances produced.

Examples :

- Breaking of glass.
- Tearing a paper

Chemical Change

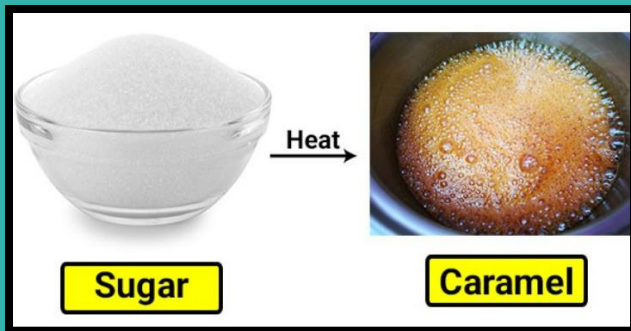
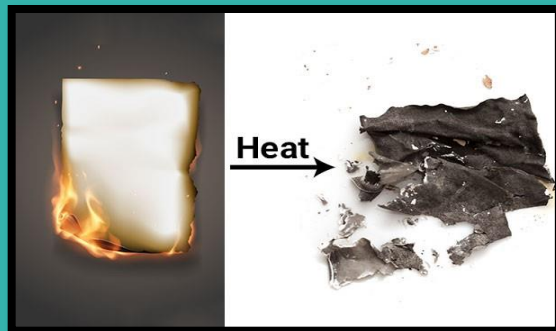
Permanent Change

New substances are produced.

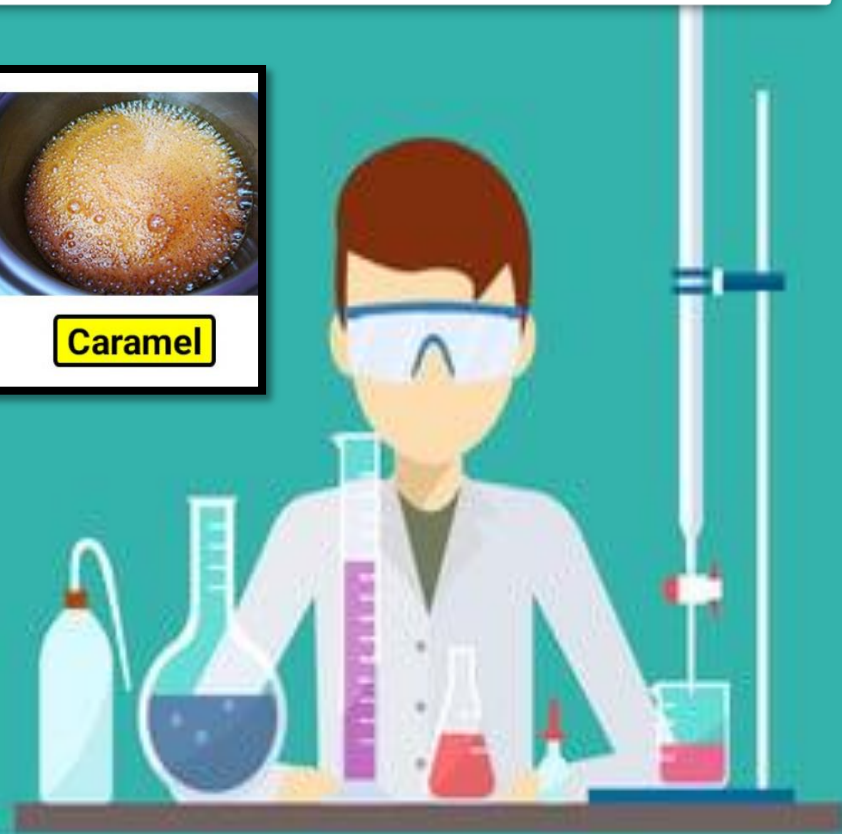
Examples :

- Cooking food.
- Burning of paper.

What is a 'Chemical Reaction'?



Whenever a chemical change occurs in our surroundings, we say a chemical reaction has take place.



What is a 'Chemical Reaction'?

Chemical Reaction - (Definition)

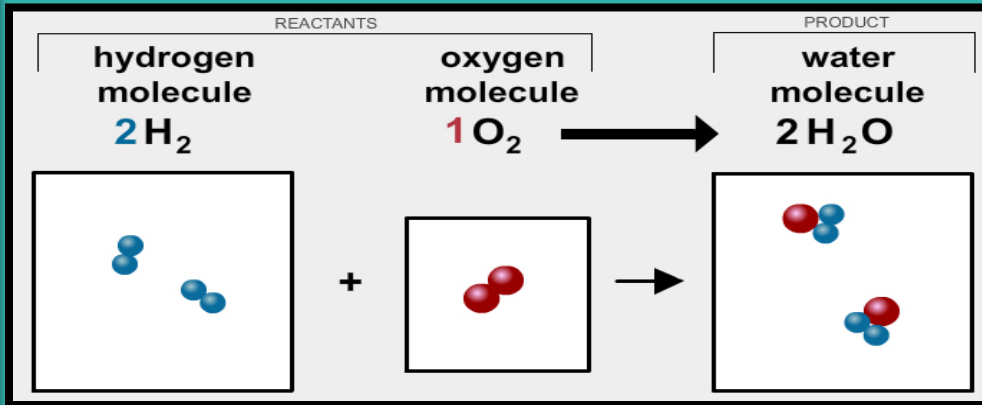
A chemical change in which two or more substances combine to form a **new substance or substances** is called a chemical reaction.



What is a 'Chemical Reaction'?

Chemical Reaction -

In a chemical reaction the **rearrangement of atom or molecules** takes place to produce **new substance**.



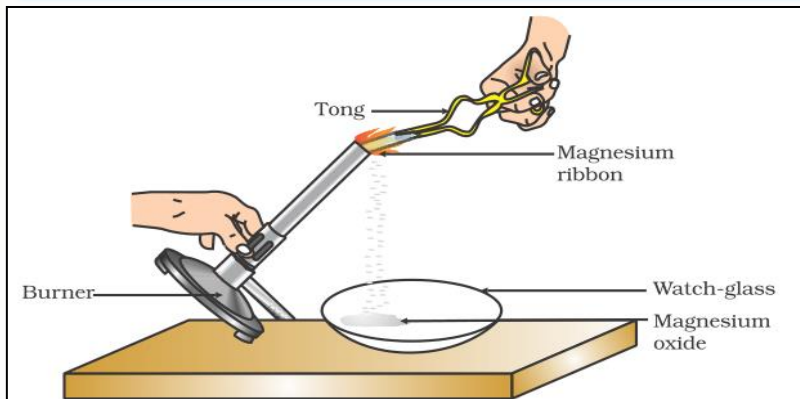
(Reactants)

(Product)



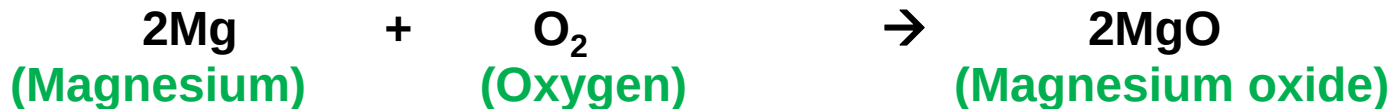
Activity – 1.1

Burning of Magnesium Ribbon is a Chemical change.



Observations :

- The magnesium ribbon burns in the presence of oxygen with a **dazzling white flame**.
- **New substance** i.e. white powder called 'magnesium oxide'.



Conclusion :

- Two reactants '**magnesium**' and '**oxygen**' reacts together to form new product '**magnesium oxide**'.
- Chemical reaction can be detected by '**Change in Temperature**'

Activity – 1.2

Reaction between Lead nitrate and Potassium Iodide



+



→



+



Lead
Nitrate

Potassium
Iodide

Lead
Iodide
(Yellow ppt)

Potassium
Nitrate

Observations :

- When '**potassium iodide**' solution is poured into '**lead nitrate**' solution, a '**double displacement reaction**' takes place.
- A **yellow precipitate** of '**lead iodide**' is formed.
- **Potassium nitrate** remains in the solution.

Activity – 1.2

Reaction between Lead nitrate and Potassium Iodide

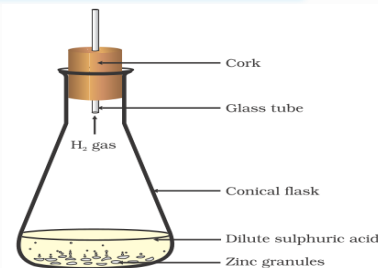


Conclusion :

- Chemical reaction between 'lead nitrate' and 'potassium iodide' can be detected by **change in the colour** of the new substances.
- Formation of solid yellow precipitate shows **change in state**.

Activity – 1.3

Reaction between Zinc and Dil. HCl produces H₂ gas



Zn
Zinc

+

2HCl
Hydrochloric Acid

→

ZnCl₂
Zinc Chloride

+

H₂
Hydrogen

Observations :

The reaction between **zinc granules** and **dilute hydrochloric acid** produces bubbles of **hydrogen gas** along with **zinc chloride** salt.

Conclusion :

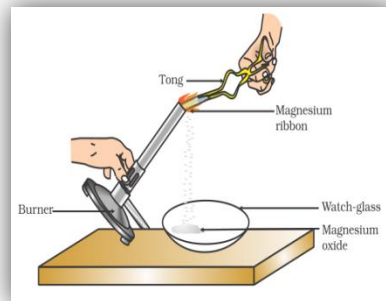
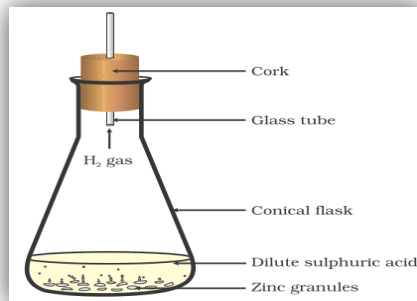
Chemical reaction between Zinc and Dil. HCl can be detected by **evolution of gas**.

Evidences that show chemical reactions has been taken place

Q. How can we determine whether a chemical reaction has taken place?

Ans: We can determine whether a chemical reaction has taken place by following observations –

1) Change in state 2) Change in colour 3) Evolution of a gas 4) Change in temperature



TO BE CONTINUED

